



What Determines AI Adoption in Companies? Mixed-Method Evidence

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ABSTRACT

Artificial Intelligence (AI) is among the emerging technologies that offer potential competitive advantages, but there is insufficient evidence regarding its use in B2B SMEs in India. This study uses empirical methods to investigate the factors that lead to the successful adoption of AI practices by B2B SMEs in India, as well as the outcomes of this adoption. A conceptual model is formulated using the technology-organization-environment (TOE) framework, which examines how AI enablers and AI readiness influence the competitive advantage with due acceptance of AI practices. To test the theoretical framework, the study used a mixed-method approach; survey data was gathered from 866 employees (managers) of SMEs. The findings suggested that all the AI enablers except perceived benefits and role clarity significantly impacted AI readiness. Further, AI ethics, the moderator of the study, was found to be significant between perceived benefits, role clarity, perceived trust, and competitive advantage.

KEYWORDS

Small and medium enterprises; AI readiness; AI ethics; mixed method

Introduction

Advanced digital technologies are bringing about significant changes in production processes and product development. This transformation, known as “Industry 4.0,” is expected to have far-reaching implications as it represents the fourth industrial revolution. In 2011, the term was introduced in Germany, and it referred to a cyber-physical system that resembled the industrial internet developed by General Electric (GE).¹ Before the emergence of Industry 4.0, businesses encountered various internal and external challenges and had to adopt advanced digital strategies to address interconnected problems.^{2,3} This required a transformative shift in manufacturing and production systems, which Industry 4.0 was specifically developed to tackle. Key components of Industry 4.0 include cyber-physical systems, artificial intelligence (AI), robotics, and the Internet of Things (IoT).^{4–6} Manufacturing companies have the potential to boost sales or optimize processes by adopting AI technologies that enable them to establish a highly flexible and efficient production system. However, despite this potential, companies are surprisingly hesitant to adopt AI, even though it could bring significant benefits.⁷ Thus far, research has offered limited explanations for the gap between the potential benefits of AI adoption and the actual adoption rates within companies.³

Prior studies showed that the Industry 4.0 paradigm requires technological development and environmental and organizational progress to facilitate the digitalization of industries.⁸ To the best of our knowledge, no studies have explored the impact of more specific organizational, technological, and environmental factors. Factors such as competitive advantage,⁶ faster decision-making,^{7,9} complexities in implementing AI,¹⁰ economic growth^{11–13} are important enablers and inhibitors during the successful adoption and integration of AI in companies. Due to the lack of research in this area, it is unclear whether findings from the broader field of organizational technology adoption can be applied to the specific context of AI adoption. However, the adoption of advanced technology by industries is impeded, as most industries are not yet prepared to implement it.¹⁰ A recent survey by McKinsey Global Institute stated that 59% of businesses are still exploring the potentialities of AI, while 40% are adopting AI.¹⁴

Previous research has examined managers’ attitudes toward AI adoption,⁹ but there is still uncertainty about their readiness to accept AI.^{15,16} Preceding studies have overlooked the competitive advantage that can be gained through the combination of resource configurations and AI, as well as the role of humans in this process.¹⁷ Additionally, the impact of AI on firms’ business strategies from a technological and environmental

perspective remains unclear and requires further investigation.^{18,19} This study aims to provide novel insights into the adoption of AI technology in SMEs. A mixed-method approach is used, to explore the use of AI in SMEs, a survey was conducted involving 866 employees (managers). Our research model was based on Tornatzky and Fleischer's (1990) Technology-Organization-Environment (TOE) framework, to investigate the companies that have integrated AI technology into their production processes and the factors that enabled their adoption.²⁰ Thus, the study formulated the following research questions:

RQ1: How do technological, organizational, and environmental prerequisites influence the adoption of AI in SMEs?

RQ2: How does AI adoption bring competitive advantage?

This paper offers a twofold contribution: first, we examine the applicability of the TOE model in identifying the technological, organizational, and environmental prerequisites that influence the adoption of AI in SMEs from a theoretical perspective. Second, our work has managerial implications that initiate the absorptive capacity of SMEs to adopt AI, the role of managers in making their firm's AI ready for competitive advantage against the backdrop of AI ethics for digital transformation in SMEs.

The present study is structured by offering related literature and the TOE framework. Followed by hypothesis development and the research design, it consists of qualitative and quantitative analysis followed by result interpretation and discussion. Finally, the study presented theoretical and managerial implications, limitations and directions for future research.

Literature review

Adoption of AI technology in small and medium-scale manufacturing firms

To remain competitive and effective, small and medium-sized manufacturing companies need to implement artificial intelligence (AI) technologies.²¹ Empirical researches have revealed that these companies have difficulty adopting AI due to a number of issues, with technology, organization, and environment being the three most important ones.²² Technology-related constraints, such as perceived complexity and expense, are important roadblocks to the adoption of AI. The high cost of investment may be prohibitive for businesses,

and they might not have the resources or knowledge to develop and operate AI systems. However, AI's relative benefits, such as increased effectiveness, productivity, and quality, might overcome these obstacles. The adoption of AI is also heavily influenced by organizational issues. AI adoption is more likely to occur in businesses with an innovative culture and an openness to experimenting with new technology.²³

A successful adoption also depends on the availability of resources like trained labor and financial capital. Studies have also emphasized the value of top management backing and a distinct adoption strategy for AI.⁷ External variables like market conditions, societal norms, and regulatory and competitive forces can also influence the adoption of AI. Companies may be more likely to employ AI, for instance, if competitors are doing so or if laws encourage its usage. Environmental constraints, such as a lack of infrastructure or a conservative business culture, might nevertheless hinder adoption.²⁴ Therefore, thorough assessment of technological, organizational, and environmental aspects as well as a desire to overcome adoption barriers are necessary for successful AI implementation in small and medium-sized manufacturing enterprises.

Technology- Organization-Environment (TOE) theory

A conceptual framework called the Technology-Organization-Environment (TOE) theory seeks to explain the factors that influence how technology is embraced and utilized in organizations. The concept was first put forward by Tornatzky and Fleischer in 1990, and it has since gained widespread acceptance in the fields of technology administration and innovation study.²⁰ According to the TOE theory, the three main elements that affect the adoption of new technologies are technological characteristics, organizational factors, and external environmental variables. This suggests that the context in which a technology is accepted and its qualities influence how it is adopted and implemented.²⁵

Several studies have applied the TOE theory to different industries and technologies, with varying results.^{8,26} For instance, the TOE theory has been used to investigate the adoption of electronic medical records (EMRs) in the healthcare sector.²⁷ According to the study, technological aspects including perceived usefulness and ease of use were the most significant determinants of EMR adoption. Additionally, it was discovered that organizational elements important to the adoption process were top management support and financial resources.

In another investigation, the retail sector's adoption of mobile payment technology was examined using the TOE theory.²⁸ The results of the study showed that organizational variables, such as top management support and training, as well as technological attributes, such as security and convenience of use, were significant predictors of the acceptance of mobile payments.

The TOE theory has drawn criticism for its shortcomings despite its widespread use. According to some academics, the notion is too general and may oversimplify the complexities of technology acceptance and implementation.²⁹ Briefly, the TOE theory provides a useful framework for comprehending the factors that influence the adoption and use of technology in organizations. Studies suggest that researchers should consider the different dimensions of technology adoption and deployment complexities while applying TOE theory.

Hypothesis development

TOE and AI adoption readiness

The readiness of an organization to adopt AI is significantly influenced by AI compatibility and benefits recognized by the organization. The success and convenience of AI adoption are heavily influenced by how well the technology integrates with current systems and procedures. The likelihood that an organization will embrace AI is increased if an AI system can be seamlessly integrated with the existing systems of the organization.³⁰ Additionally, organizations are encouraged to use AI technology by its anticipated advantages, such as higher productivity, cost savings, and improved decision-making. Employee perceptions of their responsibilities and system trust are related to how ready a company is to implement AI technology.³⁰

Employees are more likely to accept and effectively use AI if they understand how it can improve their work and have clear expectations of its use. The reliability of AI systems might impact employee trust in the technology and its accuracy.³¹ An organization's readiness for adopting AI can be increased by introducing it collaboratively, openly, and transparently with clear expectations and confidence in its capabilities. Effective data governance and quality are essential for AI adoption. Data quality promotes correctness, completeness, consistency, timeliness, and relevance while data governance ensures consistent management of data availability, usability, integrity, and security.³² Giving these criteria top priority creates a reliable data

foundation for efficient AI applications and maximum advantages. Thus, we propose:

H1: *Technological factors and AI adoption readiness are positively related to AI adoption readiness.*

H2: *Organizational Factors and AI adoption readiness are positively related.*

H3: *Environmental factors and AI adoption readiness are positively related.*

Technological factors and competitive advantage

Businesses are increasingly looking to AI to get a competitive edge in today's fast-paced commercial environment. The ability of a company to incorporate AI technologies into its systems and operations is referred to as AI compatibility. There are many advantages to this integration, including increased productivity, better decision-making, and lower expenses. However, the adoption of AI depends on both the technology's perceived value and its ability to work with an organization's current systems.³³ AI can provide a competitive edge if it is seen as a beneficial tool that can enhance productivity, cut costs, and streamline processes. When using AI to provide a competitive edge, role clarity and perceived trust are crucial factors. Employees can more successfully cooperate with the technology to produce better results when they are aware of their roles and how AI technology can be used to support them.³⁴

Additionally, employee buy-in and faith in the capabilities of the technology can be impacted by perceived trust in the AI system, which can boost adoption rates, performance, and competitive advantage.³⁵ Data governance and data quality are important variables that affect competitive advantage in the field of AI. Effective data governance ensures that data is managed, protected, and used appropriately in accordance with legal standards and corporate goals. Data accuracy, completeness, and consistency are all ensured by data quality, which is crucial for trustworthy decision-making and efficient AI algorithms.³⁶ High-quality data help organizations differentiate themselves from their rivals by giving them greater customer insights, more efficient operations, and new revenue streams. Thus, we propose:

H4: *Organizational Factors are positively related to competitive advantage.*

H5: *Organizational factors and competitive advantage are positively related.*

H6: *Environmental factors and competitive advantage are positively related.*

AI adoption readiness and competitive advantage

To preserve a competitive edge in today's swiftly evolving business environment, firms must progressively employ AI technology. Companies may raise their efficiency, accuracy, and production by successfully incorporating AI into their operations, which will lead to cost savings and more revenue.³⁷ Additionally, AI can offer insightful data on market trends and consumer behavior, empowering businesses to make wise strategic choices. The use of AI is not without its difficulties, though. To successfully integrate and apply AI, businesses must make large investments in technology, infrastructure, and talent.³⁸ As a result, a company's capacity to stay ahead of the competition is significantly influenced by its willingness to use AI. Long-term success is more likely to be attained by businesses that are better equipped to tackle the difficulties associated with adopting AI. Therefore, in order to gain a competitive edge, businesses must concentrate on increasing their AI readiness by creating strategies that successfully integrate AI into their daily operations, making investments in technology, infrastructure, and talent, and fostering an innovative and experimental culture.³⁹ By doing this, businesses can use AI to improve performance and maintain their competitiveness in the current fast-paced business environment. Thus, we propose:

H7: *AI adoption readiness and competitive advantage*

Mediation effect of AI adoption readiness

For businesses looking to obtain a competitive edge, AI has grown in importance. However, success cannot be assured by merely using AI technology. The TOE framework offers a thorough method for comprehending the different elements that affect the effective adoption of new technology. Recent studies have revealed that the relationship between TOE dimensions (i.e., technology, organization, and environment) and competitive advantage is mediated by AI readiness, or an organization's capacity to use AI technology effectively.⁴⁰ This means that a company's level of readiness for AI deployment can have a big impact on how well it can use the advantages of the technology to create a competitive edge. In order to fully realize the potential advantages of AI technology, organizations must evaluate their

degree of AI readiness while filling any gaps.⁴¹ Thus, we propose:

H8: *AI adoption readiness mediate the relationship between TOE dimensions and competitive advantage.*

Moderating effect of AI ethics

The relationship between numerous elements and competitive advantage, including as technological, organizational, environmental, and AI adoption readiness aspects, is significantly moderated by AI ethics.⁴² Organizations can prevent unfavorable outcomes, such as biased decision-making, discrimination, and privacy violations, by ensuring that AI systems are built and operated ethically and responsibly. Additionally, including ethical issues into organizational culture and decision-making can enhance employee engagement and productivity while also enhancing the organization's reputation and brand value.⁴³

AI ethics can help businesses create environmentally friendly practises that lessen their negative effects, cut expenses, and appeal to environmentally concerned customers, giving them a competitive advantage.⁴⁴ Last but not least, organizations may build trust with stakeholders, enhance their reputation, and ultimately strengthen their competitive advantage by ensuring that AI adoption is carried out in an ethical and responsible manner. Thus, we propose:

H9: *AI ethics moderate the relationship between technological factors and competitive advantage.*

H10: *AI ethics moderate the relationship between Organizational factor and competitive advantage.*

H11: *AI ethics moderate the relationship between environmental factors and competitive advantage.*

H12: *AI ethics moderate the relationship between AI adoption readiness and competitive advantage.*

Research methodology

The research analyzing AI adoption in companies using the Technology-Organization-Environment (TOE) framework was formulated based on mixed-method approaches since each type of data provides unique insights that can enrich the analysis. Qualitative data offers an in-depth understanding of the contextual factors that influence AI adoption in companies, including organizational culture, leadership support, and

employee attitudes. On the other hand, quantitative data, such as survey responses provide a broader overview of the extent and patterns of AI adoption across companies specifically the SMEs. By combining qualitative and quantitative data, researchers can triangulate their findings, strengthen their conclusions, and provide a more comprehensive and nuanced understanding of the factors that promote or hinder AI adoption in companies. Therefore, using both types of data can improve the reliability and validity of the research and enhance the applicability of the TOE framework in understanding AI adoption in companies.

Qualitative study

Netnographic method

Qualitative data were collected based on the narratives of netnographic methodology.^{45,46} Netnographic methodology is a powerful research approach that involves studying online communities and digital interactions to gain insights into consumer behavior, social trends, and cultural practices. It is particularly important in today's digital age, where online platforms have become a primary means of communication and interaction. By analyzing online conversations, behaviors, and attitudes, netnographic research can provide valuable data that can be used to inform marketing strategies, product development, and business decisions. In this study, we used non-participative observational netnography to collect data from online platforms such as discussion forums and blogs focusing on AI adoption drivers and barriers. In the first step of data collection, we identified the online communities that will be the focus of the study. This involves searching for relevant communities on social media platforms, forums, blogs and other online channels which deliberate on AI adoption challenges, drivers, outcomes, facilitating factors etc. (Table 1). Once the online communities were identified we started paying attention to the discussions, behaviors, and attitudes of the members disguising our presence. The researcher's presence was disguised so that our presence does not make the participants cautious

leading to biased opinions in the forum. Data were collected in real-time by taking screen-shots and making notes of the discussion threads. By observing and analyzing the discussions, interactions, and patterns of communication among forum members, the researcher could gain valuable insights into the AI community's beliefs, practices, and attitudes.

Analysis of qualitative data and extraction of themes

We used four-tiered inductive reasoning approach which is a comprehensive method for analyzing netnographic data in qualitative studies. The first tier involved data familiarization, where researchers immersed themselves in the data by reading through it several times to gain a deep understanding of the context and content. The second tier involved initial coding, where codes were identified and applied to the data. The third tier is focused on focused coding, where codes are refined and organized into categories, and emerging themes are identified. Finally, the fourth tier involved the development of a theoretical framework, where the researcher interprets the themes and data to develop a deeper understanding of the phenomenon being studied. Through these systematic processes, researchers identified six drivers of AI adoption in organizational categories as technological factors (perceived compatibility and perceived benefits); organizational factors (role clarity and perceived trust) and environmental factors (data governance and data quality) (Table 2) (Figure 1). Two distinct themes namely AI readiness and AI ethics also emerged from the exploration of qualitative data (Figure 1). The relationship between these overarching constructs was hypothesized based on prior literature, research knowledge and insightful discussions. In addition, the researchers assessed the reliability and validity of inductive coding to ensure the rigor and trustworthiness of the research findings. For assessing inter-coder reliability, multiple coders were incentivized to independently analyze the data and their results were compared. Through this process, we ensured that the codes and categories were consistent across different coders. Validity was assessed through a priori strategy such as

Table 1. Sources of netnographic data.

Online platform	Description of platform	Themes	Number of participants	Number of threads
Reddit>>Artificial Intelligence	Discussions of AI	Opportunities and challenges of AI adoption	18	96
Data mining forum	Data mining forum and debating platform	AI adoption environment	15	94
Youth4work	Organizational based AI debating platform	AI adoption and related organizational outcome	23	85
B-Debate	International Centre for Scientific Debate	The problems and prospects of AI in organizations	3	122
AI alignment forum	Debate on AI safety techniques	AI and organizational reengineering	25	108

Table 2. Results of inductive qualitative analysis.

Name of the factor	Sub-themes	Underlying examples from platforms	Contextual meaning	Position in the study
AI compatibility	ROI, business value, data management, integration with existing system	"But, one of the most important aspects of AI integration in organizations is compatibility. It's crucial to ensure that the AI system is compatible with the organization's existing technology infrastructure Some organizations may require more advanced AI systems, while others may just need basic ones AI integration requires careful consideration and planning. Organizations should also consider factors such as cost, data privacy, and ethical implications." "Well, for one, AI can increase efficiency. By automating certain tasks, organizations can save time and reduce errors And AI can also provide valuable insights and predictions, which can help organizations make better decisions and with the rise of chatbots and virtual assistants, AI can also improve customer service and enhance the customer experience And in some cases, AI can even help organizations identify potential risks and take proactive measures to mitigate them"	AI compatibility refers to the extent to which an AI system is aligned with the existing technology infrastructure, business processes, and organizational culture of a company	Predictor
Perceived benefits	Efficiency, productivity, customer experience, decision making, optimization	"Some people seem to think that AI will replace human workers entirely. But I think that's a bit of a misconception I think AI can help augment human capabilities and enhance productivity ... it's important for organizations to clarify the role of AI within their operations. And it's also important for organizations to ensure that their human workers have the skills they need to work alongside AI"	Perceived benefits refer to the potential advantages that an organization expects to gain from implementing AI technology. These benefits may be anticipated by leaders, employees, customers, or other stakeholders within the organization.	Predictor
Role clarity	AI roles and responsibilities, Job descriptions, communication, feedback, training	"Some people might be more willing to trust AI in certain areas, like data analysis or customer service. But in other areas, like decision-making or safety-critical tasks, people might be more hesitant to trust AI Organizations need to be transparent about how they're using AI and how it's making decisions Organizations need to make sure their AI systems are fair and inclusive I think if organizations can demonstrate that their AI systems are trustworthy and transparent, people will be more willing to trust them"	Role clarity refers to the understanding and definition of the roles and responsibilities of individuals and teams involved in the adoption, implementation, and use of AI technology within the organization	Predictor
Perceived trust	Transparency, reliability, accuracy, privacy, security	"Organizations are collecting more data than ever before, and they need to have clear policies in place for how that data is used ... AI is only as good as the data it's trained on, so organizations need to make sure their data is accurate, complete, and up to date ... Organizations need to have clear policies and procedures for data management, and they need to ensure that their AI systems are transparent and accountable And they need to ensure that their data governance policies are flexible and adaptable. As technology and regulations evolve, organizations need to be able to adapt their policies to stay in compliance and ensure that their AI systems continue to operate effectively."	Perceived trust refers to the level of confidence and trust that individuals within the organization have in the AI technology being adopted and its ability to perform as intended.	Predictor
Data governance	Data quality, data security, data ethics, data integration, data management, data privacy	"It's not just about the accuracy and bias of the data. Data quality also includes ensuring that the data is complete, consistent, and up-to-date ... This includes data cleansing, validation, and verification, as well as regular audits and monitoring It requires a cultural shift in the organization to prioritize data quality and make it part of everyone's job ... it's not just about ensuring the quality of existing data. Organizations also need to consider the quality of data they collect in the future, to ensure that it meets their needs"	Data governance refers to the process of managing and controlling the collection, storage, use, sharing, and protection of data used in AI systems.	Predictor
Data Quality	Data consistency, timeliness, relevance, integration		Data quality refers to the degree to which data used in AI systems is accurate, complete, consistent, relevant, and timely.	Predictor

(Continued)

Table 2. (Continued).

Name of the factor	Sub-themes	Underlying examples from platforms	Contextual meaning	Position in the study
AI readiness	Leadership and strategy, talent and skill, compliances, infrastructure and technology	"And it's not just about having the technical infrastructure to support AI. It's also about having the right organizational culture and mind-set They need to have a clear strategy for how they want to use AI, and how it fits into their overall business objectives . . . it's not just about the technology and strategy. It's also about having the right people in place to support AI initiatives And they need to be able to work collaboratively across different departments and functions, to ensure that AI is integrated into all aspects of the business . . ." "It's important for organizations to consider these issues and to develop ethical guidelines for the use of AI . . . It's not enough to just say that you're using AI ethically. You need to have processes in place to ensure that you're actually doing so And it's not just about compliance. Organizations need to be proactive in identifying and addressing potential ethical issues before they become problems Organizations need to be aware of the broader implications of their use of AI and to take a broader perspective on its impact . . ."	AI readiness refers to an organization's level of preparedness to effectively and efficiently adopt, implement, and integrate AI technologies into their business processes and operations. AI readiness encompasses a range of technical, organizational, and cultural factors that are critical for successful AI adoption	Mediator
AI ethics	Fairness, explainability, human oversight, responsible innovation		AI ethics refers to the moral principles and values that guide the design, development, implementation, and use of AI technologies in a responsible and ethical manner.	Moderator

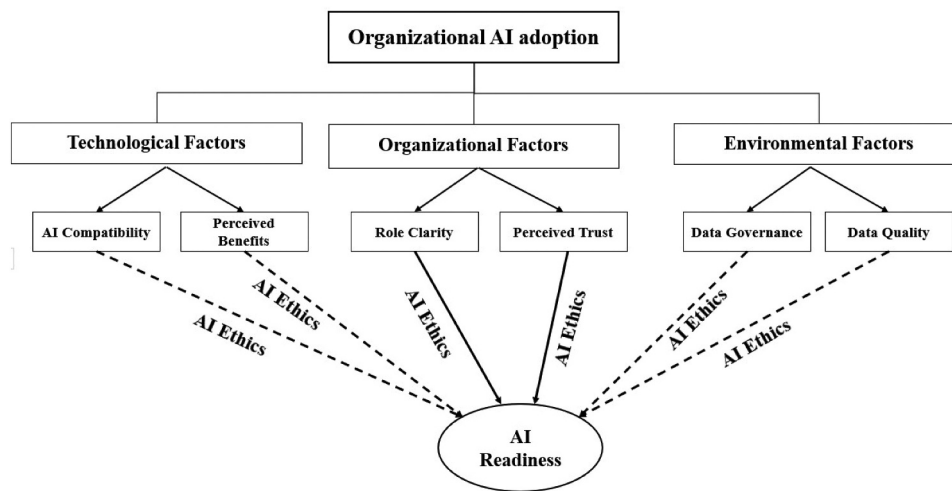


Figure 1. Results of qualitative analysis.

member checking, where the researcher shares their findings with some of the participants to ensure that they accurately reflect their experiences and perspectives, and triangulation, where multiple sources of data from other platforms were also used to corroborate the findings. This process of ensuring reliability and validity of the study was crucial to ensure that the research findings are trustworthy and can be applied to other contexts.

The results of the qualitative study using inductive qualitative analysis aim to generate insights and understanding from the data without any preconceived hypotheses. However, the themes that emerge from the analysis provided us useful clues for generating hypotheses and testing relationships in quantitative framework. For instance, in the present study, the themes explored can be comprehensively categorized as technological factors, organizational factors and environmental factors using which a hypothesis could be that each of these factors have positive relationship with AI readiness and competitive advantage, and this could be tested by collecting quantitative data pertaining to AI adoption in companies. Therefore, while inductive qualitative analysis does not rely on preconceived hypotheses, the themes identified can provide a starting point for generating hypotheses and testing relationships in quantitative research.

Quantitative study

Methods and materials

The current study, which is descriptive in nature, illustrates the different obstacles to AI adoption in the Indian manufacturing industry. Based on the TOE framework, the necessary data was gathered by an Indian

market research firm. We were unable to compile a comprehensive list of businesses that have explored or incorporated AI into their business processes because AI adoption is still in its infancy in India. Therefore, we employed judgmental sampling to select our respondents for this study, which involved recruiting manufacturers of SMEs who are aware of AI and related disruptions. Manager-level employees of the SMEs were administered structured questionnaire via an online survey. In order to improve generalization of results, the research agency was previously recommended to cover a variety of industries across SMEs with a wide demographic. We added two screening questions to the study instrument, “Are you aware of the potentials of AI in your industry” and “Does your company fit within the SME sector,” as a precaution to look for inconsistencies. The study excluded those who responded “no” to either of the questions. Eventually, we had 866 valuable data, which we used in additional multivariate analysis. The analytical framework consists of covariance based structural equation modeling using IBM AMOS graphics. We compiled the test of mediation and measurement model in IBM AMOS whereas moderation was carried out through Process Macros by Andrew F Haynes. Confirmatory factor analysis was used to assess the psychometric qualities of the measurement model. After confirming the measures of reliability and validity, structural coefficients depicting the direct and indirect path were compiled. After confirming the presence of mediation, moderation role was also tested in the proposed model.

Measures

The constructs used in the study were all adopted from the prior literatures and established scales. However, the

statements were suitably modified to choose the current context. All the constructs were measured in Five-Point Likert scale. Four items measuring AI compatibility and four-items indicating perceived benefits were modified from the works of various researches.^{47,48} The works of researches⁴⁹ were referred to develop the measures of role clarity. Three items measuring perceived thrust were adapted from the work of various researches on AI.⁵⁰ Three items from the work of various researches⁵¹ were used to measure data governance, whereas three-items were used to indicate data quality.⁵² AI readiness was measured using three items adapted from various sources.⁵³ Three items indicating competitive advantage were adapted from the works of various researches.⁵⁴ Finally, AI ethics measured using three items were adapted from the works of various sources⁵⁵ (Table 3).

Reliability and validity assessment

The psychometric properties assessed using confirmatory factor analysis were found satisfactory (CMIN/DF = 3.076, GFI = 0.916 AGFI = 0.895, CFI = 0.966, TLI = 0.960, RMSEA = 0.049).⁵⁶ Convergent validity was pronounced through AVE values and found that all the values are higher than 0.50.⁵⁷ It was also found that composite reliability values were decent and satisfy the threshold level of 0.70. Divergent validity was ensured using Fornell-Larcker criterion. It was observed that square root of AVE values were higher than the intercorrelation between the constructs depicting divergent validity in the measurement model⁵⁸ (Table 4). Further, measures of skewness and kurtosis were used to assess the parametric nature of data and found that the collected data confirm the properties of

Table 3. Results of confirmatory factor analysis.

Construct	Statements	Label	Loadings
AI Ethics	AI respects privacy, and there will be no leak of personal and confidential information	AIE1	0.82
	AI is safe and never discriminate	AIE2	0.893
	AI ensures sensitivity to different values and human moralities	AIE3	0.899
AI Compatibility	AI is compatible with most aspects of the organization	AIM1	0.942
	AI suits the organization style	AIM2	0.916
	Using AI suits the way organization intends to function	AIM3	0.891
	AI technology suits my organizational needs	AIM4	0.82
Perceived Benefits	AI helps in proper decision making	PEB1	0.933
	A lot of alternatives can be examined through AI	PEB2	0.923
	The strategies of AI are helpful to the organization	PEB3	0.909
	It is worthwhile to invest in AI for the benefits it will give	PEB4	0.898
Role Clarity	I have clear, planned goals and objectives for my job	ROC1	0.893
	I receive assignments that are with my training and capability	ROC2	0.877
	I just have the right amount of work to do	ROC3	0.778
	Explanation of what needs to be done is clear to me	ROC4	0.792
Perceived Trust	The AI systems are trustworthy	PET1	0.944
	I do not doubt the honesty of AI systems	PET2	0.916
	The AI systems are competent and effective	PET3	0.838
Data Governance	I feel AI data meets the necessary compliance	DGA1	0.921
	AI data management does not have any bias	DGA2	0.864
Data Quality	AI demands authorization to access sensitive data	DGA3	0.892
	The quality of data are excellent	DQA1	0.875
	Data of the organization is handled with care	DQA2	0.902
	Business risks are less when data quality management is proper	DQA3	0.881
AI Readiness	I believe that the main data sets of the organization are of high quality and reliable	DQA4	0.802
	I understand AI integrates different domains of organization	AIR1	0.93
	I understand AI-based systems are more precise and structured	AIR2	0.732
	AI functions are highly versatile and broadly applicable	AIR3	0.781
Competitive Advantage	Proper utilization of AI benefits in achieving superior business performance	CAD1	0.886
	Proper utilization of AI results in employee productivity	CAD2	0.851
	Successful AI-Human collaboration optimizes efficiency and complementary capabilities	CAD3	0.829

Table 4. Results of construct validity.

	CR	AVE	MSV	MaxR(H)	CAD	AIM	PEB	ROC	PET	DGA	DQA	AAR	AIE
CAD	0.891	0.732	0.368	0.894	0.856								
AIM	0.940	0.798	0.440	0.950	0.568	0.893							
PEB	0.954	0.839	0.259	0.955	0.364	0.509	.916						
ROC	0.903	0.700	0.238	0.913	0.488	0.403	.213	0.837					
PET	0.928	0.811	0.381	0.940	0.584	0.617	.455	0.407	.900				
DGA	0.922	0.797	0.382	0.926	0.597	0.588	.381	0.488	.603	0.893			
DQA	0.923	0.750	0.440	0.928	0.607	0.663	.495	0.474	.606	0.618	0.866		
AAR	0.858	0.670	0.312	0.901	0.494	0.534	.379	0.364	.509	0.529	0.559	0.819	
AIE	0.904	0.759	0.279	0.911	0.342	0.497	.480	0.214	.525	0.463	0.528	0.460	0.871

normal distribution.⁵⁹ Finally, tolerance and variance inflated factor measures were used to partial out the issues of multicollinearity in the regression model.⁶⁰

Participants

It is important to provide context and a possible explanation for the demographic results found in the analysis of AI adoption in Indian SMEs. It was observed that majority of the participants are male (73.2%) and belong to 45–55 years age group (43.1%). Individuals of this age group may have a better understanding of the potential benefits of AI adoption and the associated risks, making them more likely to participate in the study. The study also found that majority of the respondents were earning 120k–150k per month (45.8%) comprising the typical middle and upper-middle-class professionals in India. It is possible that respondents in this income range may have a better understanding of AI adoption and may be more likely to have access to the necessary resources and infrastructure to adopt AI in their SMEs.

Common method variance

The vulnerabilities of common method variance were controlled by following the step-by-step approaches recommended by prior literatures.⁶¹ First, we have used mixed methodology for operationalizing this study. The results of qualitative study were used to hypothesizing and empirical validation in the quantitative methods. Second, Harman single-factor test results were found satisfactory and no generic factor was observed in the explored factor structure. Due care

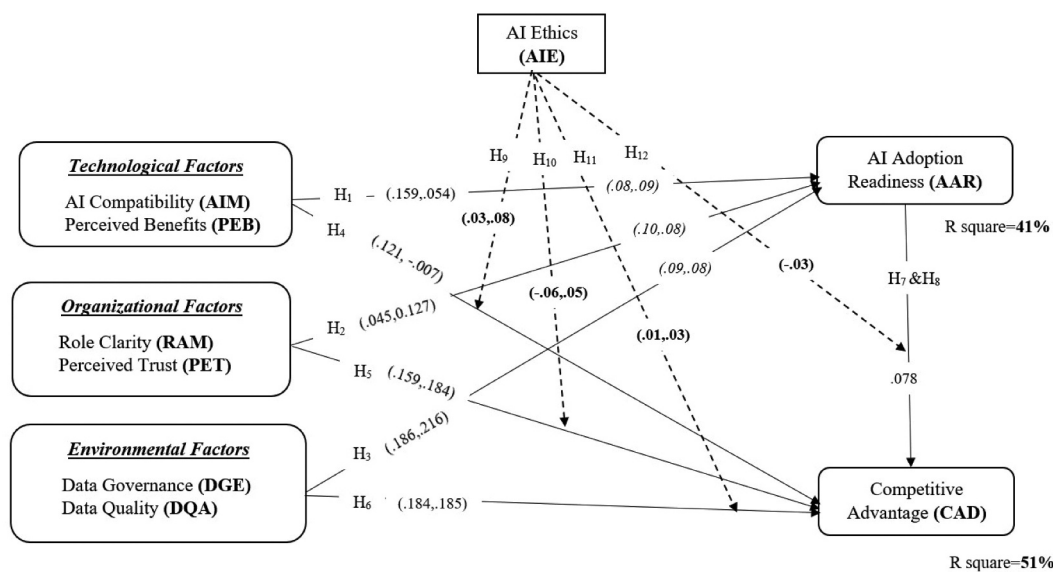
was taken while administering the questionnaire to reduce item ambiguity and sequential biases. Detailed instruction for filling the questionnaire together with randomization of items reduced the probabilities of respondent biases. As a measure of prudence, the inter-correlation between constructs was used and found to be per threshold limit (Table 4). As a last process, we introduce a single unobserved latent factor into the model and examine for any significant changes in the structural coefficients. It was found that the presence of the latent factor does not produce any notable changes to the existing coefficients reinforcing the absence of common method biases in this study.

Results of hypotheses testing

The results of a mediation and moderation analysis examining the direct, indirect, and conditional influence of TOE dimensions on competitive advantage of the company in the context of AI adoption are detailed in this session. The results are shown in Figure 2.

Mediation analysis

Results showing the summary of series of direct effect were presented in (Table 5). It was found that AI compatibility has significant impact on AI readiness whereas perceived benefits failed to produce significant impact leading to partial acceptance of H₁ ($\beta_1 = .159$, $\beta_2 = .054$). Similarly, among the organizational factors, only perceived trust had significant impact on AI readiness



*Coefficients in italic parenthesis indicate (mediation effect); coefficients in bold parenthesis indicate (moderation effect) and coefficients in non-bold, non-italic parenthesis indicate (direct effect) in the model

Figure 2. Results of hypotheses testing.

Table 5. Summary of direct path coefficients.

Path	Estimate	S. E	C.R	P	Hypothesis	Supported
AAR $\leftarrow$ AIM	.159	.041	3.395	.001	H1	Yes
AAR $\leftarrow$ PEB	.054	.029	1.475	.140		No
AAR $\leftarrow$ ROC	.045	.038	1.220	.223	H2	No
AAR $\leftarrow$ PET	.127	.042	2.852	.004		Yes
AAR $\leftarrow$ DGA	.186	.037	4.079	.001	H3	Yes
AAR $\leftarrow$ DQA	.216	.048	4.363	.001		Yes
CAD $\leftarrow$ AIM	.121	.039	2.780	.005	H4	Yes
CAD $\leftarrow$ PEB	-.007	.028	-.205	.837		No
CAD $\leftarrow$ ROC	.159	.037	4.599	.001	H5	Yes
CAD $\leftarrow$ PET	.184	.041	4.432	.001		Yes
CAD $\leftarrow$ DGA	.184	.036	4.315	.001	H6	Yes
CAD $\leftarrow$ DQA	.185	.046	3.990	.001		Yes
CAD $\leftarrow$ AAR	.078	.039	2.036	.042	H7	Yes

signifying partial acceptance of H_2 ($\beta_3 = .045$, $\beta_4 = .127$). H_3 is accepted since both the environmental factors had significant impact on AI readiness ($\beta_5 = .186$, $\beta_6 = .216$). It was observed that among the technological factors, AI compatibility had significant influence on competitive advantage partially accepting H_4 ($\beta_7 = .121$, $\beta_8 = .007$). Both H_5 ($\beta_9 = .159$, $\beta_{10} = .184$) and H_6 ($\beta_{11} = .185$, $\beta_{12} = .185$) were prima facie accepted since both dimension of organizational and environmental factors significantly influences organizational competitive advantage. Finally, it was found that organizational AI readiness significantly leads to competitive advantage accepting H_7 ($\beta_{13} = 0.78$). The results of hypothesis testing the mediating role of AI readiness in the conceptual model are given below (Table 6).

The results of mediation revealed the presence of partial mediating role of AI readiness accepting H_8 . It was found that AI readiness significantly mediates all the direct path between predictors and organizational competitive advantage. This means that the ability of an organization to adopt and utilize AI technologies effectively can enhance its competitive advantage in the market. Further, the moderating role of AI ethics in amplifying the competitive advantage of companies due to AI adoption is illustrated below (Table 7).

Moderation analysis

Tests of moderation through AI ethics revealed the presence of significant moderation effect between perceived barriers, role clarity, perceived trust resulting in partial acceptance of H_9 and full acceptance H_{10} . It was also observed that no significant moderation was found between environmental factors and outcome variable as well as AI readiness and competitive advantage rejecting H_{11} and H_{12} . The research results indicate that AI ethics significantly moderates the relationship between Technology-Organization-Environment (TOE) dimensions and competitive advantage (Figure 3). This suggests that organizations need to consider ethical considerations in the deployment of AI technologies in order to leverage the full potential of these technologies and gain a competitive advantage. Furthermore, the study highlights the importance of considering the wider context in which AI technologies are deployed. The results of moderation proved that organizations that prioritize ethical considerations in the deployment of AI technologies are more likely to gain a competitive advantage, as they will be able to leverage the full potential of these technologies while minimizing any negative impacts.

Table 6. Summary of mediation analysis.

Path	Direct Effects	ULCI	LLCI	Indirect Effects	ULCI	LLCI	Sig.	Mediation	Hypo
AIM $\rightarrow$ AAR $\rightarrow$ CAD	.36	.31	.41	.08	.06	.11	Yes	Partial	H_8
PEB $\rightarrow$ AAR $\rightarrow$ CAD	.20	.14	.25	.09	.07	.12	Yes	Partial	
ROC $\rightarrow$ AAR $\rightarrow$ CAD	.34	.28	.41	.10	.07	.12	Yes	Partial	
PET $\rightarrow$ AAR $\rightarrow$ CAD	.39	.33	.44	.08	.06	.11	Yes	Partial	
DGA $\rightarrow$ AAR $\rightarrow$ CAD	.43	.37	.49	.09	.06	.12	Yes	Partial	
DQA $\rightarrow$ AAR $\rightarrow$ CAD	.43	.37	.49	.08	.05	.11	Yes	Partial	

Table 7. Summary of moderation analysis.

Path	β	SE	t	P	LLCI	ULCI	Moderation	Hypothesis
AIM $\rightarrow$ CAD	.03	.02	1.47	.14	-.01	.08	No	H_9
PEB $\rightarrow$ CAD	.08	.02	3.37	.00	.03	.13	Yes	
ROC $\rightarrow$ CAD	-.06	.03	-2.13	.03	-.11	-.01	Yes	H_{10}
PET $\rightarrow$ CAD	.05	.02	1.98	.05	.00	.09	Yes	
DGA $\rightarrow$ CAD	.01	.02	.61	.54	-.03	.06	No	H_{11}
DQA $\rightarrow$ CAD	.03	.02	1.35	.18	-.01	.08	No	
AAR $\rightarrow$ CAD	-.03	.03	-.93	.35	-.09	.03	No	H_{12}

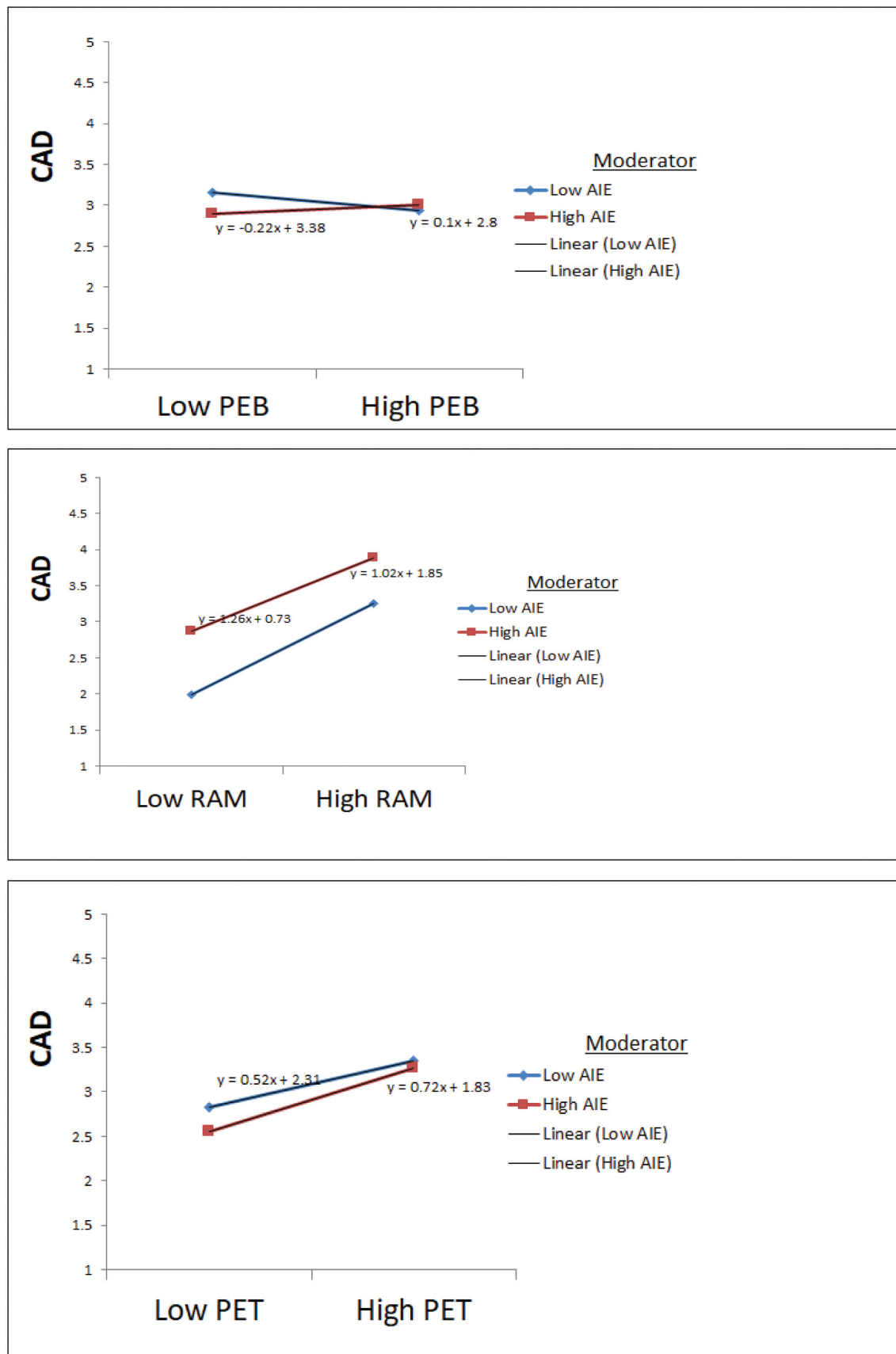


Figure 3. Parsimonious measure of moderation.

Discussions

The research examining AI adoption in Indian SMEs using the TOE (Technology-Organization-Environment) framework provides insights into the factors that influence the adoption of AI in SMEs. The study found that the adoption of AI in SMEs in India is influenced by three main categories of factors: technology-related factors, organizational-related factors, and environmental-related factors as informed by the results of netnographic methodology and inductive qualitative analysis. The hypothesized relationship between the theoretical constructs revealed many path-breaking results. Examining the direct impact of TOE factors on AI readiness revealed that AI compatibility, perceived trust, data governance and data quality had significant positive impact on AI influence whereas two of the predictors namely perceived benefits and role clarity do not have significant influence over AI readiness (H_1 , H_2 , H_3). Organizational AI readiness refers to the ability of an organization to effectively adopt and utilize artificial intelligence (AI) technologies to improve its operations, decision-making processes, and overall performance. One of the key factors that can impact an organization's AI readiness is the compatibility of its existing systems and infrastructure with AI technologies. If an organization's systems and infrastructure are not compatible with AI, it may be challenging for the organization to integrate AI technologies effectively. However, if an organization's systems and infrastructure are designed to be compatible with AI, it can enable easier adoption and implementation of AI technologies, thus improving the organization's AI readiness. Another important factor that can impact an organization's AI readiness is the perceived benefits of AI. If an organization understands the potential benefits of AI, such as increased efficiency, cost savings, improved decision-making, and better customer service, it is more likely to invest in and adopt AI technologies. This perception of the benefits of AI can also drive the development of strategies to support the organization's AI readiness. Effective data governance can help ensure that the organization's data is accurate, accessible, and usable, which is critical for successful AI implementation. This is because AI relies on high-quality data to learn and make accurate predictions or decisions. Finally, the quality of an organization's data has a significant effect on its AI preparedness. For AI to make trustworthy predictions or choices, it needs access to high-quality data. In light of this, it is crucial for businesses to put data quality first by implementing data cleaning, standardization, and validation procedures. Organisational and environmental factors were

shown to have a substantial impact on competitive advantage, whereas only AI compatibility from the technological elements determined to have a statistically significant beneficial impact (H_4 , H_5 , H_6). Consequently, AI readiness also positively influence competitive advantage of companies in AI adoption. Our result supported the previous findings that artificial intelligence (AI) has the potential to significantly impact an organization's competitive advantage.⁶² By automating repetitive tasks and augmenting human decision-making, AI can improve accuracy, speed, and consistency, leading to better business outcomes. Clear role definitions and responsibilities for AI implementation are critical to ensure that employees understand the value and purpose of AI within the organization. With clear role definitions, employees are more likely to embrace the technology and leverage its benefits to their advantage. Building trust among employees toward AI is important to ensure its adoption and effective utilization. Effective data governance is crucial for ensuring the accuracy, completeness, and consistency of data used in AI models. Organizations that can manage their data assets well can leverage AI to gain insights and make better decisions, giving them a competitive advantage over others. Finally, organizations that invest in data quality initiatives, such as data cleaning, normalization, and standardization, can improve the accuracy and reliability of their AI models. This, in turn, can lead to better decision-making, more efficient processes, and cost savings, ultimately providing a competitive advantage.

The results of mediation analysis found that AI readiness plays a partial-mediating role between all the TOE factors and competitive advantage (H_8). The mediating role of AI readiness between TOE dimensions and organizational competitive advantage can be explained by the fact that an organization's readiness to adopt AI can significantly influence its ability to leverage the benefits of AI and gain a competitive advantage. This, in turn, can lead to improved efficiency, productivity, and cost savings, ultimately providing a competitive advantage. Therefore, our findings highlight the importance of investing in AI readiness initiatives to fully realize the benefits of AI adoption. Finally, significant moderation effect of AI ethics was observed between perceived benefits, role clarity, perceived trust and outcome variable (H_9 , H_{10}); however, no significant moderation effect was observed between the environment factors, AI readiness and outcome variable (H_{11} , H_{12}). These findings

of moderation analysis have high relevance in AI adoption. Our results proved that impact of AI compatibility on organizational competitive advantage is moderated by AI ethics, which ensures that the AI system is developed and deployed in ways that align with the organization's values and ethical principles. Second, impact of AI ethics on the relationship between role clarity and organizational competitive advantage ensures that employees' roles and responsibilities are aligned with ethical principles. Third, the moderating impact of AI ethics on the relationship between perceived trust and organizational competitive advantage ensures that AI systems are developed and used in ways that align with ethical principles and values.^{63,64} The moderating impact of AI ethics on the relationship between organizational competitive advantage and the other variables ensures that the benefits of AI adoption are achieved without compromising ethical principles.²² The results of the present study are congruent with the past literatures discussing the roadmap for AI adoption in organizations. Our study focusing exclusively on SMEs provide innovative approach regarding the process of AI adoption.⁶⁵ Comparing with the prior literatures, our findings offer more congruent and consistent picture about the state of AI adoption in SMEs. The current findings have numerous practical and theoretical implications also.

Implications of the study

Theoretical implications

The research examining the state of AI adoption using the Technology-Organization-Environment (TOE) framework with AI ethics as a moderator and AI readiness as a mediator offers several important theoretical contributions. Firstly, it highlights the importance of considering the dynamic and complex interplay between technological, organizational, and environmental factors in determining the adoption of AI. By using the TOE framework, the study provides a comprehensive and integrative perspective on the adoption of AI that takes into account multiple factors that influence the decision-making process. Secondly, the study's incorporation of AI ethics as a moderator indicates the significance of ethical issues in influencing the spread of AI technologies. Specifically, the study found that businesses with a higher emphasis on ethics were more inclined to invest in artificial intelligence. This discovery implies that ethical principles can function as a driver of AI adoption and emphasizes the need for their incorporation into organizational decision-making processes.

Finally, the study's use of AI readiness as a mediator offers important insights into the adoption process of AI technologies. Specifically, the study suggests that organizations with greater readiness for AI adoption are more likely to adopt AI technologies, and that this readiness is influenced by a range of technological, organizational, and environmental factors. This finding underscores the importance of considering organizational preparedness when implementing AI technologies, and suggests that organizations may need to invest in a range of capabilities to successfully adopt AI.^{66,67}

Overall, the study's use of the TOE framework blended with moderator and mediator provides a comprehensive and integrative perspective on the adoption of AI, highlighting the importance of considering a range of factors that influence the adoption process. The study's findings offer important theoretical implications for both researchers and practitioners interested in understanding and facilitating the adoption of AI technologies.

Managerial implications

The study has several managerial implications; firstly, the study's results can inform the development of managerial strategies to guide future acceptance of AI technology within organizations. In order to promote the adoption of AI technology within manufacturing and production firms, it is essential for managers to be transparent and communicate the benefits of this technology to all stakeholders. Additionally, designers and developers of AI technology should be mindful of creating user-friendly systems that are not overly complicated. It is important for managers to emphasize the practical benefits of AI adoption within the organization to ensure acceptance and support from all stakeholders. The study also found that perceived trust has a significant impact on AI readiness.

To increase the usefulness of AI in an organization, managers should provide appropriate training to employees to enhance their competency and expertise. This can increase their willingness to adopt AI and make it easier for the organization to implement. Managers should also consider readiness as a vital predictor of adoption success, and therefore, clearly define the strategy for adoption, secure strong and committed sponsorship for resource mobilization, provide active support for personnel and informed about the benefits of AI adoption, ensure that the data used by AI are reliable and actionable, and maintain effective communication throughout the adoption process. The study showed role clarity is not significantly impacted on AI readiness, which shows the personnel were not given a proper clarity on their works to be done.

Small and medium-sized firms often have limited resources to invest in new technologies. Therefore, managers should carefully consider the costs and benefits of adopting new technologies and select those that align with their business goals and strategies and ensure that employees have the necessary skills and training to effectively use new technologies. Managers should create a culture that encourages innovation and experimentation, while also providing clear guidelines and processes for implementing new technologies. Small and medium-sized firms operate in a constantly changing business environment, which can impact their ability to adopt and implement new technologies, thus managers should stay up-to-date on industry trends, legal and regulatory implications, technological advancements and be prepared to adapt their strategies.^{66,67}

Limitations and future direction

The Technology-Organization-Environment (TOE) framework is a widely recognized framework that provides a holistic view of the factors that influence the adoption of new technologies by organizations, including small and medium-sized enterprises (SMEs). However, despite its usefulness, the research focusing on the TOE framework for examining the AI readiness and adoption barriers in Indian SMEs have certain limitations. The sampling method may not have taken into account the heterogeneity of the SMEs in India, such as differences in industry sectors, geographic location, and size. Therefore, the findings of this study may not be generalizable to all Indian SMEs. Additionally, online surveys can have response bias, where participants may not provide honest or accurate responses due to social desirability bias or other factors. Thirdly, the TOE framework used in this research may not have captured all the factors that influence AI adoption in Indian SMEs. For instance, cultural factors, such as trust in technology and risk aversion, may play a critical role in the adoption of AI, but may not have been adequately addressed by the framework. Therefore, the framework's limited scope may have led to an incomplete understanding of the barriers to AI adoption in Indian SMEs. Finally, the research did not examine the actual adoption of AI technologies by Indian SMEs, and the findings are based on the perceptions of the managers. The study did not examine the actual implementation of AI or the challenges that SMEs faced during implementation. Therefore, the study's findings may not accurately reflect the real situation of AI adoption in Indian SMEs.

The potential research directions that could build on this work include examining the role of individual-

level factors in AI adoption. The current study focused on organizational-level factors, but individual-level factors such as employee attitudes and perceptions about AI, AI literacy, and training could play a crucial role in AI adoption. While the study highlighted the moderating role of AI ethics in the relationship between TOE factors and AI adoption, future research could explore the impact of regulatory factors such as data privacy laws, intellectual property rights, and cybersecurity regulations on AI adoption in Indian SMEs. The study focused on AI adoption in Indian SMEs. However, a comparative study could be done with other countries to understand the similarities and differences in the factors influencing AI adoption. AI adoption has the potential to impact employment, and future research could explore the impact of AI adoption on job displacement and job creation in Indian SMEs. Overall, the current research provides a solid foundation for future research that can help advance our understanding of AI adoption in Indian SMEs and inform the development of practical frameworks for implementing ethical AI practices.

Conclusions

The Technology-Organization-Environment (TOE) framework was utilized in the study to examine the factors impacting the adoption of artificial intelligence (AI) in small and medium-sized enterprises (SMEs) in India. The study found that several technological, organizational, and environmental factors significantly impacted how AI became prevalent in SMEs. Various characteristics, including AI compatibility, perceived trust, data governance, and data quality, were used to predict the readiness for AI adoption.⁶⁷ Contrarily, perceived advantages and role clarity had no appreciable influence. Clear role definitions and duties, effective data governance, and high-quality data all proved essential for successful AI implementation. The study also discovered that TOE factors and organizational competitive advantage were slightly mediated by AI readiness. The relationship between organizational competitive advantage and AI adoption has also been shown to be significantly moderated by AI ethics. The study's theoretical and practical ramifications help us understand AI adoption in SMEs more thoroughly, and its findings are crucial for SMEs intending at employing AI.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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